

Unitrak User Guide

NCSSM Amtrak Shuttle Tracker

Unitrak is a small website that helps NCSSM students and Community Coordinators (CCs) keep track of the Amtrak shuttle out of Durham station (station code DNC). Students can see every upcoming train, sign up for a shuttle, cancel, or mark themselves as boarded. CCs see a live roster grouped by train and can mark each student as picked up or no show, leave notes, and mark a van as departed.

This guide has two parts. Part 1 lists every command needed to start the project on your computer. Part 2 walks through each page in the order you would visit it and tells you what every button does.

Part 1: Starting The Project

Do these steps once, in order, in the same terminal window.

Step 1. Install Node.js and Docker Desktop

Install both of these with their default options:

- Node.js LTS (version 20 or higher) from <https://nodejs.org>
- Docker Desktop from <https://www.docker.com/products/docker-desktop>

Open Docker Desktop after installing and wait until the whale icon in your menu bar or system tray stops animating. That means Docker is ready.

Open a new terminal and check that Node is installed:

```
node --version
npm --version
```

Step 2. Open a terminal in the project folder

Unzip the submission, then open a terminal inside the folder that contains package.json.

- Mac: right click the folder in Finder and pick "New Terminal at Folder."
- Windows: hold Shift, right click the folder, and pick "Open in Terminal" (or "Open PowerShell window here").
- Linux: open your terminal and run `cd /path/to/the/folder`.

Confirm you are in the right place. This should print package.json with no error:

```
ls package.json
```

Step 3. Start Redis with one Docker command

With Docker Desktop running, run this in the terminal:

```
docker run --name unitrak-redis -p 6379:6379 -d redis:7
```

The first time, Docker downloads the image, which takes about a minute. Check that it is running:

```
docker ps
```

If Docker says the container already exists from a previous run, start it instead:

```
docker start unitrak-redis
```

Step 4. Make the .env file

Copy the template. On Mac or Linux:

```
cp .env.example .env
```

On Windows PowerShell:

```
Copy-Item .env.example .env
```

Open the new env any text editor and look for these three values

```
NEXTAUTH_SECRET="any-random-string-you-choose"
```

```
GOOGLE_CLIENT_ID="your-google-client-id"
```

```
GOOGLE_CLIENT_SECRET="your-google-client-secret"
```

Run `openssl rand -base64 32` in your terminal to generate a secure NEXTAUTH_SECRET and use it.

GOOGLE_CLIENT_ID and GOOGLE_CLIENT_SECRET come from the Google Cloud Console. To get them:

- Go to <https://console.cloud.google.com> and create or select a project.
- Navigate to APIs & Services then go to Credentials and click Create Credentials then go to OAuth client ID.
- Set the application type to Web application.
- Under Authorized redirect URIs, add `http://localhost:3000/api/auth/callback/google`.
- Click Create, then your Client ID and Secret will appear. Paste them into .env.

Step 5. Install and create the database

Run these three commands in order. Wait for each to finish:

```
npm install
npm run prisma:generate
npx prisma db push
```

The first downloads dependencies, the second builds the database client, the third creates the SQLite database file (dev.db). You may see warnings, but there should be no errors.

Step 6. Start the app

```
npm run dev
```

After a few seconds the terminal prints "Ready" and a link to <http://localhost:3000>. Leave that terminal open and visit that link in any browser.

Common Problems

- `NODE_MODULE_VERSION` error: run `npm rebuild better-sqlite3`, then try again.
- Port 3000 already in use: run `PORT=3001 npm run dev` (on Windows PowerShell use `$env:PORT=3001; npm run dev`).
- Docker says it cannot connect: Docker Desktop is not running. Open it and wait for the whale to settle.
- Student dashboard shows "No trains loaded": open <http://localhost:3000/api/trains?sync=1> once to force the schedule download.
- Want to start over fresh: stop the app with Ctrl-C, delete dev.db, run `npx prisma db push`, then `npm run dev`.

Coming Back Later

To stop the app, press Ctrl-C in the terminal running `npm run dev`. To stop Redis too, run `docker stop unitrak-redis`.

Next time, just open Docker Desktop, then run these two commands in the project folder:

```
docker start unitrak-redis
npm run dev
```

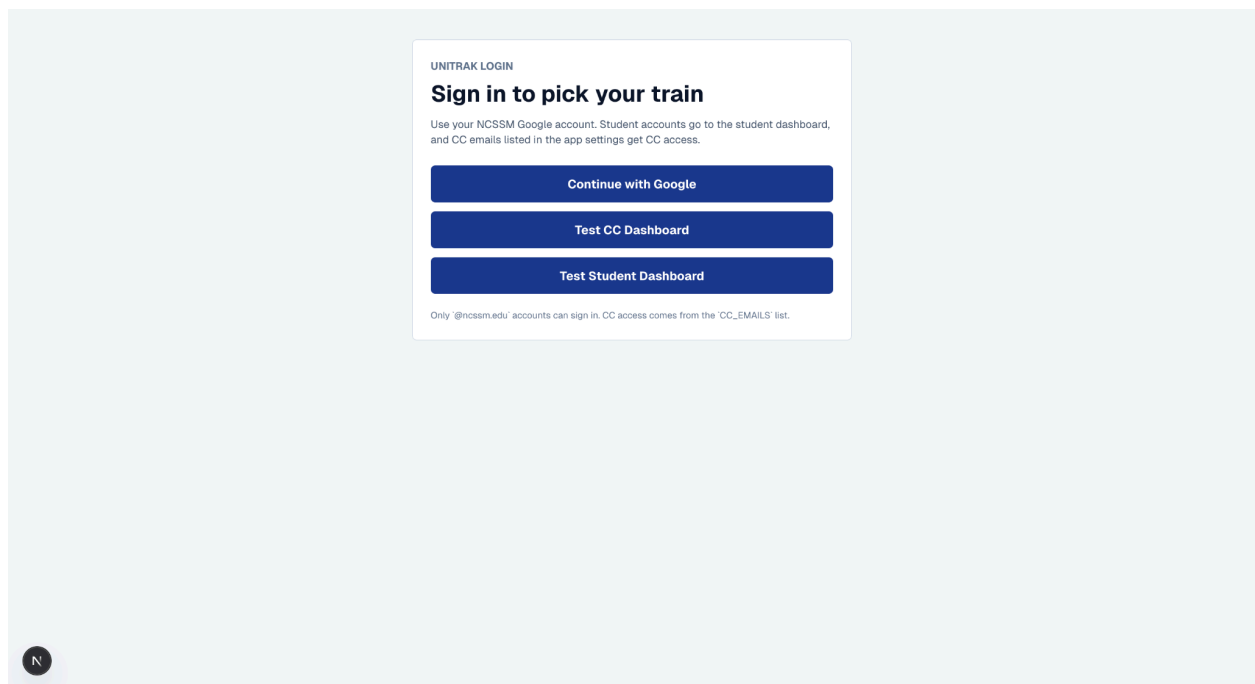
Part 2: Using The App

From here on, everything happens at <http://localhost:3000> in your browser.

The Login Page

Going to <http://localhost:3000> sends you to the login page at `/login`. Every user starts here. You will see:

- A small label that reads UNITRAK LOGIN and the title "Sign in to pick your train."
- A blue button labelled Continue with Google. This is the real login. It uses NCSSM Google accounts.
- A second outlined button labelled Test CC Dashboard. This is a temporary button just for testing.



Important: the real Google button only works if a Google OAuth client id and secret are in `.env`. We did not include our school's OAuth credentials in this submission, so for grading please use the Test CC Dashboard button. It signs you in as a sample CC user without needing a Google account. From the CC dashboard you can then click Student view in the top right to also see the student side.

Clicking "Test CC Dashboard"

Clicking the Test CC Dashboard button takes you to `/cc`, which sets a cookie, creates a fake user named Test CC (email `test.cc@ncssm.edu`, role `cc`), and redirects you to the CC dashboard. This button is enabled by `ALLOW_CC_TEST_LOGIN` in `.env` (set to `true` by default for grading).

CC DASHBOARD

Live shuttle roster

Signed in as test.cc@ncssm.edu

Student viewHistorySign out

Date: 2026-05-19Refresh

Train 77 to Charlotte

Leaves DNC at 3:47 PM · Piedmont · 2 students

Mark van departed

Van departed at 3:22 PM

Pranav Mahesh

mahesh26p@ncssm.edu

no_show

Picked upNo-show

Test Student

test.student@ncssm.edu

picked_up

Picked upNo-show

Train 74 to Raleigh

Leaves DNC at 5:51 PM · Piedmont · 0 students

Mark van departed

Van departed at 3:22 PM

No students signed up.

Train 76 to Raleigh

Leaves DNC at 7:40 PM · Piedmont · 0 students

Mark van departed

No students signed up.

The Student Dashboard

The student dashboard lives at /student. From the CC dashboard, click Student view in the top right to reach it.

STUDENT DASHBOARD

Unitrak

Signed in as test.cc@ncssm.edu

HomeCC viewSign out

Upcoming Durham Departures

Rough flow: choose the train, then CCs see you in their roster.

Train 77 to Charlotte

Tue, May 19, 3:47 PM · Piedmont · GTFS

I boardedCancel

Current: active

Train 74 to Raleigh

Tue, May 19, 5:51 PM · Piedmont · GTFS

Sign up

Train 76 to Raleigh

Tue, May 19, 7:40 PM · Piedmont · GTFS

Sign up

Train 79 to Charlotte

Tue, May 19, 8:03 PM · Carolinian · GTFS

Cancel

Current: boarded

Train 78 to Raleigh

Tue, May 19, 10:37 PM · Piedmont · GTFS

Sign up

Train 71 to Charlotte

Wed, May 20, 7:02 AM · Piedmont · GTFS

Sign up

Train 80 to New York

Wed, May 20, 9:37 AM · Carolinian · GTFS

Sign up

Train 77 to Charlotte

Wed, May 20, 3:47 PM · Piedmont · GTFS

Sign up

Train 74 to Raleigh

Sign up

The header at the top shows a small STUDENT DASHBOARD label, the big title Unitrak, and "Signed in as <email>." In the top right are three buttons: Home, CC view (only shown if you have the CC role), and Sign out.

Below the header is a card titled Upcoming Durham Departures, listing every train at DNC over the next 72 hours. Each row shows:

- A bold line: Train <number> to <destination>.
- The scheduled departure in Eastern Time, the route name, and the data source.
- On the right, the action buttons. They change depending on your current signup state.

The action buttons work like this:

- If you are NOT signed up: one blue Sign up button. Clicking it reserves your spot.
- If your signup is active: a blue I boarded button and an outlined Cancel button. "Current: active" appears under them.
- If you already marked yourself boarded or a CC picked you up: only Cancel is shown, with the current status under it.

Train 76 to Raleigh Tue, May 19, 7:40 PM · Piedmont · GTFS	I boardedCancel Current: active
Train 79 to Charlotte Tue, May 19, 8:03 PM · Carolinian · GTFS	Cancel Current: boarded

Below the trains is a second card called My Trip History. It lists every signup you have ever made, newest first. Each row shows the train number, the status (active, boarded,

picked_up, cancelled, or no_show), the departure time, and the destination.

My Trip History	
Train 80 - boarded	Thu, May 21, 9:37 AM to New York
Train 76 - boarded	Wed, May 20, 7:40 PM to Raleigh
Train 80 - boarded	Wed, May 20, 9:37 AM to New York
Train 71 - boarded	Wed, May 20, 7:02 AM to Charlotte
Train 76 - active	Tue, May 19, 7:40 PM to Raleigh
Train 79 - boarded	Tue, May 19, 8:03 PM to Charlotte
Train 77 - active	Tue, May 19, 3:47 PM to Charlotte

Clicking a row opens the trip detail page (/student/trips/<id>), which is a bigger view of one signup with the same Cancel and I boarded buttons at the bottom.

[Back to dashboard](#)

Train 76

Piedmont to Raleigh

Departure
Tue, May 19, 7:40 PM

Current signup status
active

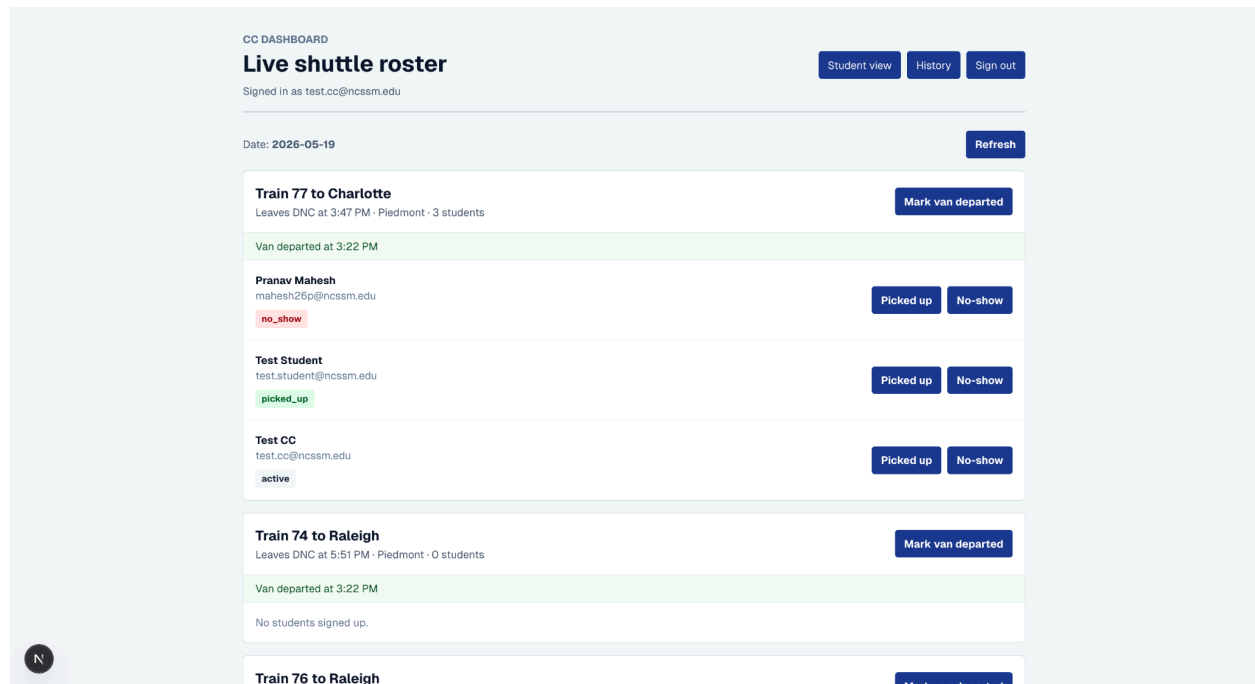
I boarded

Cancel

Current: active

The CC Dashboard

After the Test CC Dashboard button (or a real CC Google login), you are on /cc. This is the page CCs use on shuttle day.



The header shows a small CC DASHBOARD label, the title Live shuttle roster, and your email. In the top right are Student view, History, and Sign out. Just below the header is the current date (in Eastern Time) and a Refresh button. The page also auto refreshes through a live event stream, so manual refreshes are rarely needed.

Below the date is a list of train cards, one per upcoming train. Each card has:

- A header bar with "Train <number> to <destination>," the leave time, route name, and how many students signed up.
- A dark Mark van departed button on the right of that header bar. Clicking it opens a prompt asking "CC notes for this shuttle run?" Type a note (for example, "left at 5:42, light traffic") or leave it blank and press OK. A green "Van departed at <time>" banner appears under the header.
- One row per signed up student. Each row shows the student's name, their email, a coloured status pill, a blue Picked up button, and an outlined blue No-show button.

Train 77 to Charlotte
Leaves DNC at 3:47 PM · Piedmont · 3 students

Mark van departed

Van departed at 3:22 PM

Pranav Mahesh mahesh26p@ncssm.edu no_show	Picked up No-show
Test Student test.student@ncssm.edu picked_up	Picked up No-show
Test CC test.cc@ncssm.edu active	Picked up No-show

Click Picked up when a student boards the van. The box turns green. Click No-show if a student signed up but never came; the box turns red. If a card has no students signed up, it shows "No students signed up."

Because of the live event stream, signups, cancellations, and pickups from other tabs (or other CCs' browsers) update on this page on their own within a second.

CC History And Audit Log

Clicking History in the CC dashboard header takes you to /cc/history. The page has two cards:

- Shuttle Runs: one row per van that has been marked departed, with the train, departure time, and any notes you saved.
- Audit Log: one row per recorded action across the whole app (signup created, boarded, picked up, no show, cancelled, van departed, schedule synced). Each row shows the action, the time, the actor's email (or "system"), the target, and any metadata.

[Back to CC dashboard](#)

History and Audit Log

Rough record of shuttle departures and signup changes.

Shuttle Runs

Train 80 - boarding

Thu, May 21, 9:37 AM

Train 76 - boarding

Wed, May 20, 7:40 PM

Train 80 - boarding

Wed, May 20, 9:37 AM

Train 71 - boarding

Wed, May 20, 7:02 AM

Train 76 - boarding

Tue, May 19, 7:40 PM

Train 79 - boarding

Tue, May 19, 8:03 PM

Train 74 - departed

Tue, May 19, 5:51 PM - departed Tue, May 19, 3:22 PM

Train 77 - departed

Tue, May 19, 3:47 PM - departed Tue, May 19, 3:22 PM

Train 72 - departed

Wed, May 13, 1:02 PM - departed Wed, May 13, 9:40 AM

N

Signing Out

Every dashboard has a Sign out button in the top right. Clicking it clears both the Google session and the test CC cookie, and returns you to the login page. From there you can use Test CC Dashboard again to come back in.

Thanks for trying Unitrak.